Research activity

Order and repetition in counting — 30 min.

Six distinct students volunteer to organise a debate.

1. Two students are chosen to form a pair with no roles assigned. How many pairs can be formed?
2. A chair and a secretary are appointed, who must be two different students. How many assignments of the two roles are possible?
3. Three successive speeches are planned. The same student may speak several times. How many programmes of three speeches are possible?
4. Start again with only four students, denoted A, B, C and D. Write out all the pairs and all the assignments of the two roles. Then explain why there are 64 programmes of three speeches, without listing them.
5. For each of the three ways of choosing, state whether order matters and whether the same person may be chosen several times.